

VAIBHAV CHHABRA

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SUMMARY

Senior Detection Engineer and Threat Researcher with 10+ years of experience building behavioral detection systems across cloud-native and Linux environments. Specialized in telemetry correlation, adversary emulation, and anomaly-driven analytics to improve detection precision and reduce operational noise. Proven track record designing container-based attack testbeds and multi-source telemetry pipelines (CloudTrail, Kubernetes Audit, XDR) to strengthen real-world detection efficacy. Strong Python, Linux, AWS, and Docker expertise focused on scalable, reliable security engineering.

SKILLS

Security Engineering & Detection Infrastructure: Linux internals, container observability, detection pipelines, EKS, cron persistence, escape detection, telemetry correlation, real-time alerting, detection triage automation

Threat Research & Detection Engineering: Adversary behavior modeling, OPSEC evasion analysis, MITRE ATT&CK, Sigma, Falco, Procmon, EDR/XDR, IOC clustering, behavioral anomaly analytics

Programming & Automation: Python, Shell, PowerShell, C/C++, REST APIs, detection automation, log parsing, workflow orchestration

Visualization & Communication: Plotly, Matplotlib, Seaborn, Kibana, interactive dashboards, technical reporting

Cloud & DevOps: Docker, Kubernetes, AWS, Git, CI/CD, containerized testbed orchestration

Machine Learning & Analytics: Scikit-learn, Pandas, NumPy, PCA, UMAP, HDBSCAN, RBM, VAE, anomaly detection, statistical scoring (z-scores), clustering and drift analysis

Malware Analytics: Malware telemetry analysis, VirusTotal automation, YARA integration, PE analysis, Exploit PoC validation, Protocol analysis

EXPERIENCE

Security ML & Anomaly Detection | Jun 2025 – Present
Independent Researcher

[Project: Behavioral Anomaly Detection in the Tor Network](#)

- Designed an unsupervised ML pipeline for behavioral anomaly detection in Tor relay populations.
- Built autoencoders for feature compression and projected latent representations with UMAP.
- Clustered relays with HDBSCAN and modeled centroid movement, density shifts, role transitions, and uptime resets.
- Applied statistical scoring methods including z-scores and density analysis.
- Visualized telemetry patterns through interactive dashboards and behavioral maps.
- Resulted in a patent-pending anomaly detection framework and research publications on arXiv.

Trend Micro, Austin, TX | Mar 2022 - Jun 2025
Threat Researcher

- Researched Linux container and cloud attack techniques including escape-to-host, privilege escalation, and fileless persistence.
- Built adversary emulation testbeds in Kubernetes (EKS) to simulate TTPs and generate detection telemetry.
- Orchestrated ART pipelines and leveraged LLMs to analyze failures, generate remediations, and improve detection coverage.
- Authored Sigma rules for Linux and Kubernetes techniques mapped to MITRE ATT&CK and integrated into Trend Micro XDR.
- Correlated Linux audit logs, AWS CloudTrail, and EDR/XDR telemetry to reduce false positives and improve signal quality.
- Partnered with detection and engineering teams to close coverage gaps and improve production efficacy.

Trend Micro, Austin, TX | Sep 2013 - Mar 2022
Security Researcher

- Authored network IPS filters to detect vulnerability exploitation and protocol abuse.
- Reproduced attack PoCs (e.g., Metasploit modules) in lab environments for detection signature development.
- Built pattern-based detection filters aligned with exploit payloads and lateral movement techniques.
- Delivered high-performance production filters protecting enterprise environments in real time.
- Developed tooling integrating VirusTotal feeds, YARA processing, Portable Executable (PE) analysis, Trend
- Locality Sensitive Hash (TLSH), and heuristic analysis to enrich malware intelligence.

PATENTS

- *Network Traffic Pattern Based Machine Readable Instruction Identification*
U.S. Patent No. [US10757029B2](#)
- *System and Method for Geometric Manifold-Based Anomaly Detection in Networks*
[U.S. Provisional App. No. 64/034,500](#) (Patent Pending)

PUBLICATIONS

- Chhabra, V. (2026). Latent Geometry as a Structural Monitor. [arXiv:2605.20391](#)
- Chhabra, V. (2026). Drift as Deformation. [Zenodo: 18463659](#)

EDUCATION

University of Oklahoma – BS, Electrical Engineering (Minor: Physics)

University of Miami – MS, Music Engineering Technology

- Applied wavelet packet decomposition for scalable audio feature extraction.